

Learning Objectives

- Demonstrate an understanding of the energetics of phase transitions by creating heating/cooling curves
- Calculate heats of phase transition, $q = n \Delta H_{\text{phase transition}}$
- Calculate heat transfer within one phase, $q = C_s m \Delta T$
- Calculate total heat absorbed or released over a temperature range that includes a phase transition.
- Analyze and sketch phase diagrams

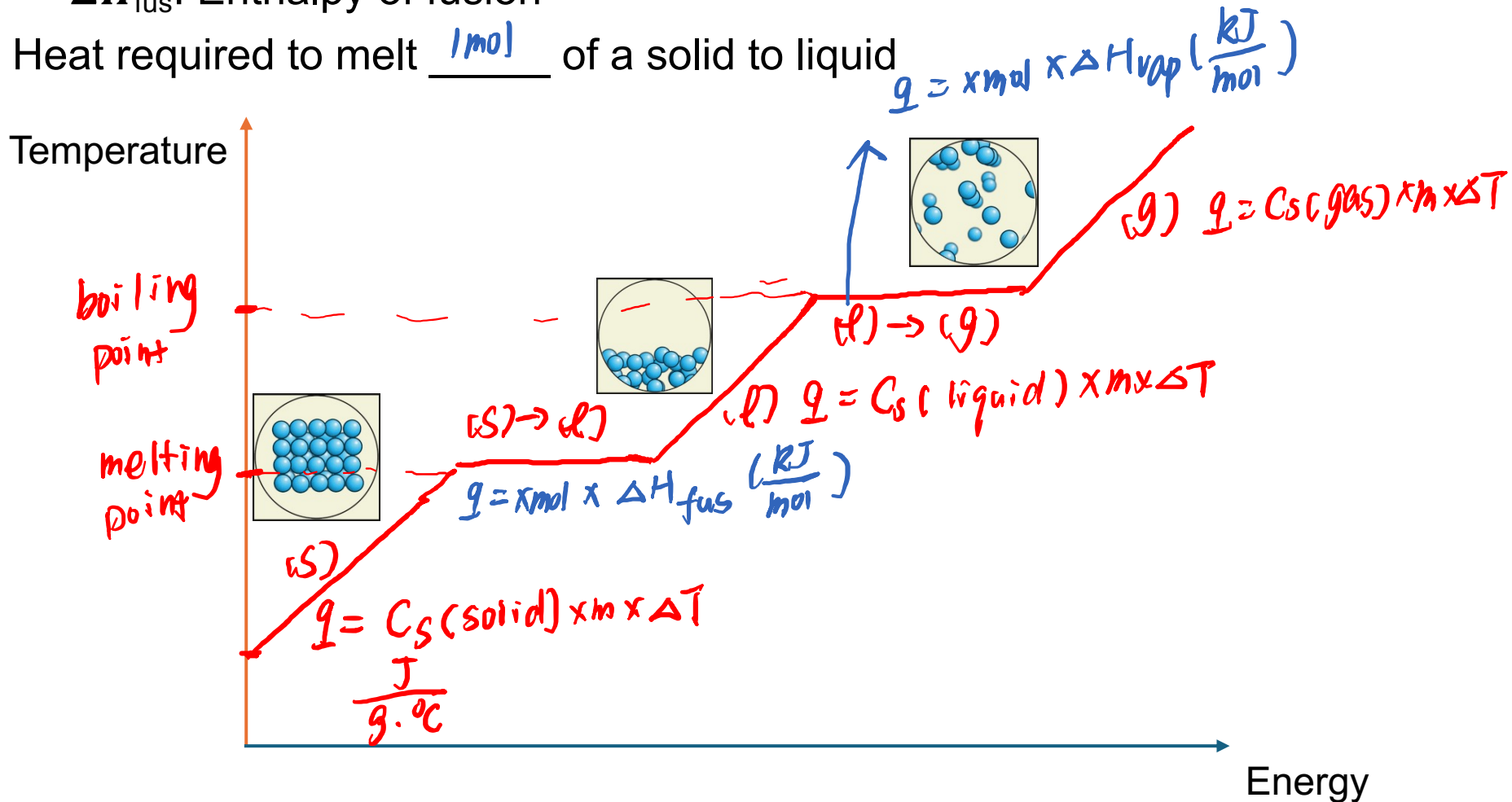
Heating Curve

- ΔH_{vap} : Enthalpy of evaporation

Heat required to evaporate 1 mol of a liquid to gas

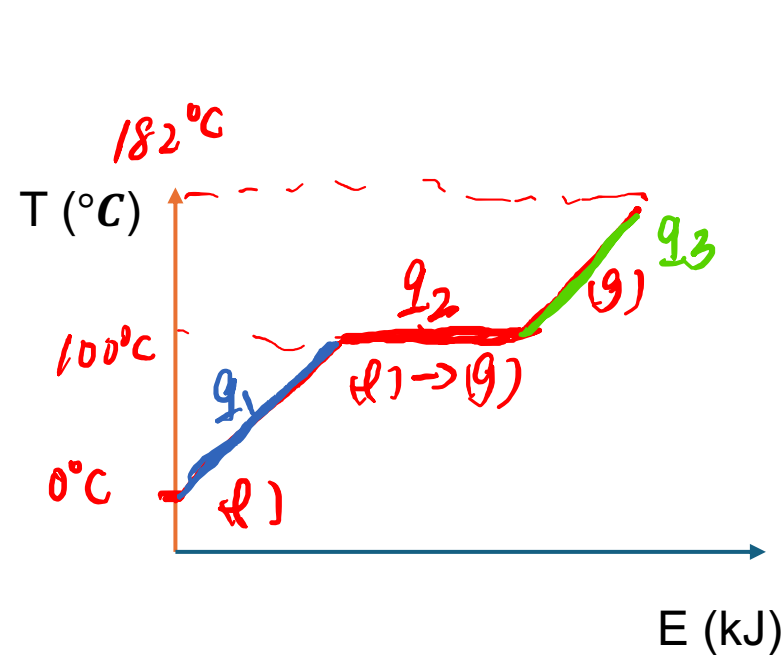
- ΔH_{fus} : Enthalpy of fusion

Heat required to melt 1 mol of a solid to liquid



Calculating ΔE from Heating Curve

Calculate the amount of energy (in kJ) needed to heat 346 g of liquid water from 0°C to 182°C . Assume that the specific heat of water is $4.184 \text{ J}/(\text{g} \cdot ^\circ\text{C})$ over the entire liquid range and that the specific heat of steam is $1.99 \text{ J}/(\text{g} \cdot ^\circ\text{C})$. ΔH_{vap} for water is 40.79 kJ/mol . Water: b.p. 100°C m.p. 0°C



$$q_1 = 4.184 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}} \times 346 \text{ g} \times (100^\circ\text{C} - 0^\circ\text{C}) \times \frac{1 \text{ kJ}}{10^3 \text{ J}}$$

$$= 145 \text{ kJ}$$

$$q_2 = 346 \text{ g} \times \frac{1 \text{ mol}}{18.00 \text{ g}} \times \frac{40.79 \text{ kJ}}{1 \text{ mol}} = 783 \text{ kJ}$$

$$q_3 = 1.99 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}} \times 346 \text{ g} \times (182^\circ\text{C} - 100^\circ\text{C}) \times \frac{1 \text{ kJ}}{10^3 \text{ J}}$$

$$= 56.5 \text{ kJ}$$

$$q_{\text{total}} = 145 \text{ kJ} + 783 \text{ kJ} + 56.5 \text{ kJ}$$

$$= \boxed{985 \text{ kJ}}$$

The heating process is exothermic / endothermic and the sign of q should be ____.

Calculating ΔE from Heating Curve

Calculate the amount of energy (in kJ) needed to heat 346 g of liquid water from 0°C to 182°C . Assume that the specific heat of water is $4.184 \text{ J}/(\text{g} \cdot ^\circ\text{C})$ over the entire liquid range and that the specific heat of steam is $1.99 \text{ J}/(\text{g} \cdot ^\circ\text{C})$. ΔH_{vap} for water is 40.79 kJ/mol .

Segment 1: Heating liquid water from 0°C to 100°C .

$$q_1 = ms\Delta t$$
$$= (346 \text{ g}) \left(\frac{4.184 \text{ J}}{\text{g} \cdot ^\circ\text{C}} \right) (100^\circ\text{C} - 0^\circ\text{C})$$
$$= 1.45 \times 10^5 \text{ J} = 145 \text{ kJ}$$

Segment 2: Phase change at 100°C .

$$q_2 = n\Delta H_{\text{vap}}$$
$$= (346 \text{ g}) \left(\frac{1 \text{ mol}}{18.02 \text{ g}} \right) (40.79 \text{ kJ})$$
$$= 783 \text{ kJ}$$

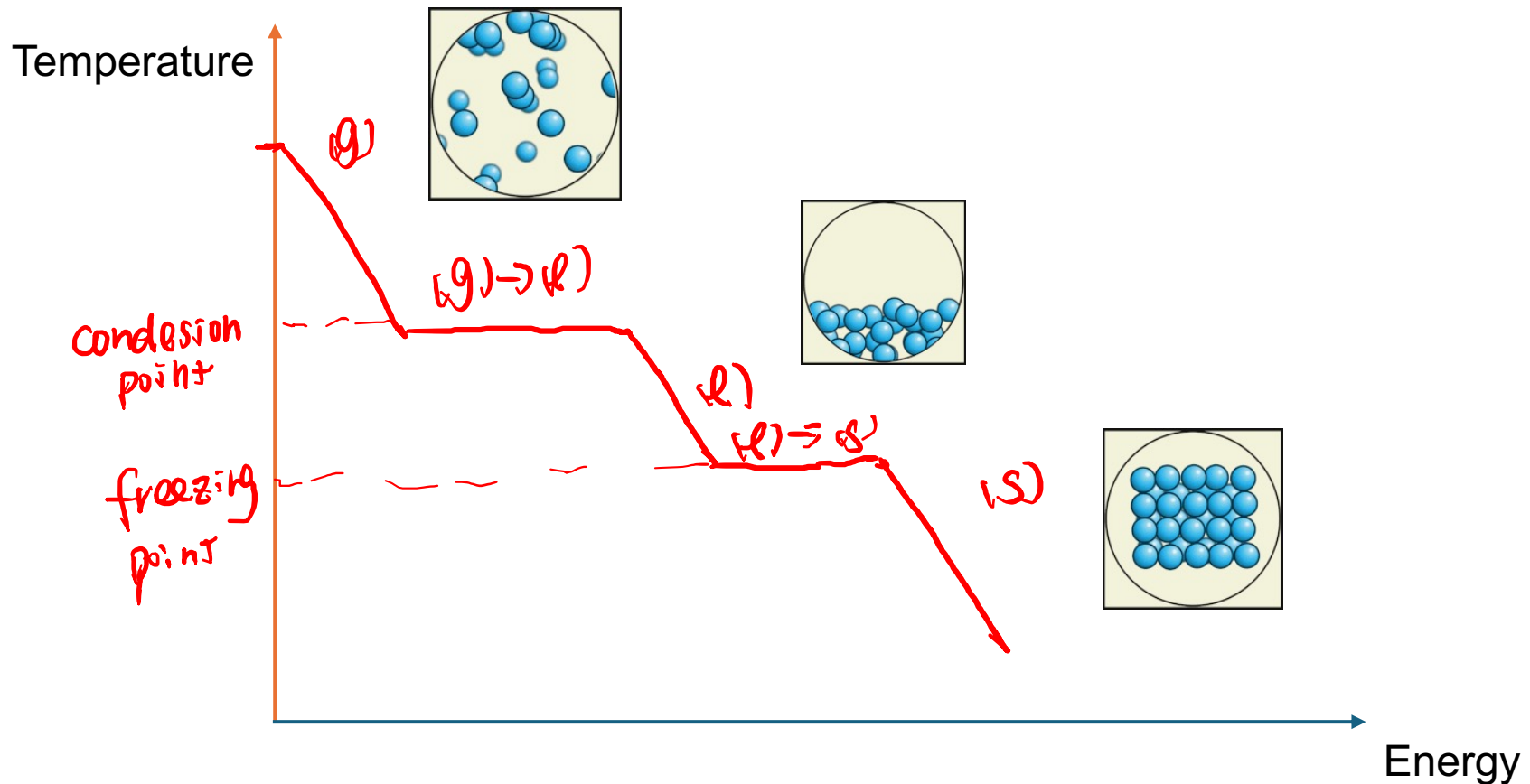
Segment 3: Heating steam from 100°C to 182°C .

$$q_3 = ms\Delta t$$
$$= (346 \text{ g}) \left(\frac{1.99 \text{ J}}{\text{g} \cdot ^\circ\text{C}} \right) (182^\circ\text{C} - 100^\circ\text{C})$$
$$= 5.65 \times 10^4 \text{ J} = 56.5 \text{ kJ}$$

$$q_{\text{total}} = q_1 + q_2 + q_3 = 145 \text{ kJ} + 783 \text{ kJ} + 56.5 \text{ kJ} = 985 \text{ kJ}$$

Cooling Curve

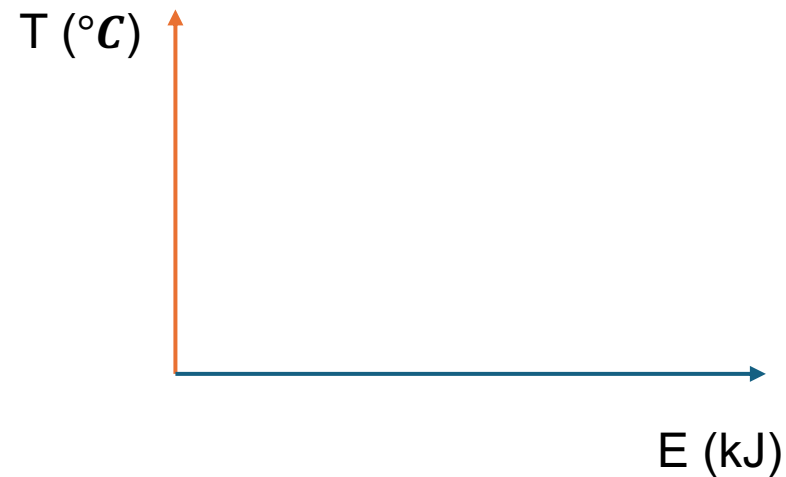
- ΔH_{cond} : Enthalpy of condensation = — ΔH_{vap}
- ΔH_{frez} : Enthalpy of freezing = — ΔH_{fus}



The cooling process is exothermic / endothermic and the sign of q should be ____.

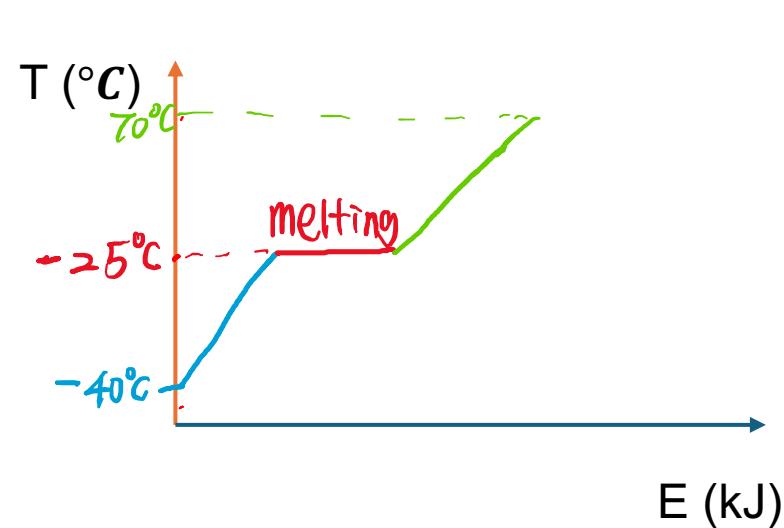
Your Turn!

How much energy in kJ is needed to change 150.g of a substance from a solid at -40.0°C to a liquid at 70.0°C . The melting point of the substance is -25°C . Enthalpy of fusion is $1200. \text{ J/g}$. Enthalpy of vaporization is $3000. \text{ J/g}$. $C_{\text{s (solid)}} = 3.00 \text{ J/g}^{\circ}\text{C}$ and $C_{\text{s (liquid)}} = 6.20 \text{ J/g}^{\circ}\text{C}$.



Your Turn!

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$$q_1 = \frac{3.00 \text{ J}}{\text{g}^{\circ}\text{C}} \times 150. \text{ g} \times (-25^{\circ}\text{C} - (-40^{\circ}\text{C})) \times \frac{1 \text{ kJ}}{10^3 \text{ J}} = 6.75 \text{ kJ}$$

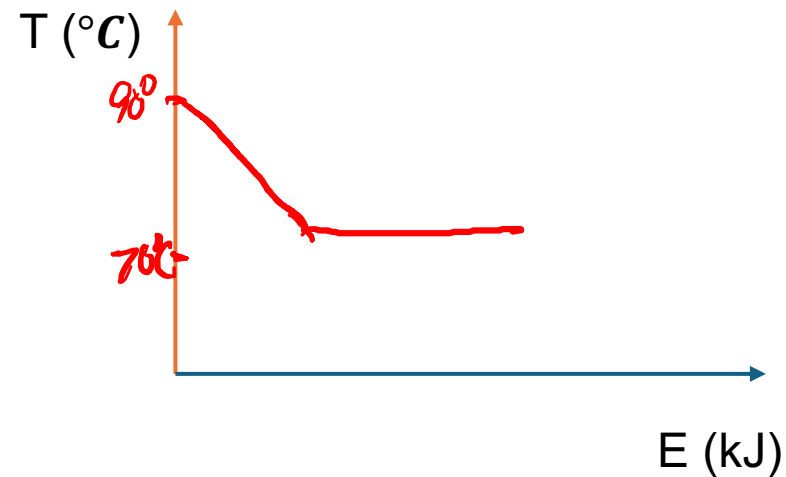
$$q_2 = 1200. \frac{\text{J}}{\text{g}} \times 150. \text{ g} \times \frac{1 \text{ kJ}}{10^3 \text{ J}} = 180 \text{ kJ}$$

$$q_3 = \frac{6.20 \text{ J}}{\text{g}^{\circ}\text{C}} \times 150. \text{ g} \times (70^{\circ}\text{C} - (-25^{\circ}\text{C})) \times \frac{1 \text{ kJ}}{10^3 \text{ J}} = 88.5 \text{ kJ}$$

$$q_{\text{total}} = 275 \text{ kJ}$$

Your Turn!

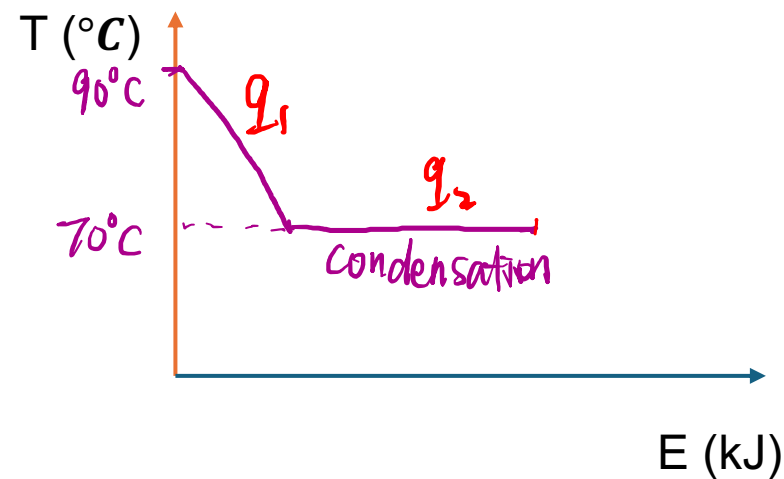
Given a sample of ethanol (mm = 46.07 g/mol), boiling point = 70 °C, $C_{s \text{ gas}} = 1.3 \text{ J/g}^\circ\text{C}$ and $\Delta H_{\text{vap}} = 88 \text{ kJ/mol}$. Calculate heat, in kJ, for the 5.0 g of alcohol from 90 °C through the process of complete condensation to 70 °C. Include a heating/cooling curve.



Your Turn!

Given a sample of ethanol (mm = 46.07 g/mol), boiling point = 70 °C, $C_{s \text{ gas}} = 1.3 \text{ J/g}^\circ\text{C}$ and $\Delta H_{\text{vap}} = \underline{88 \text{ kJ/mol}}$. Calculate heat, in kJ, for the 5.0 g of alcohol from 90 °C through the process of condensation to 70 °C. Include a heating/cooling curve.

$$q = \underbrace{5.0 \text{ g} \times 1.3 \frac{\text{J}}{\text{g}^\circ\text{C}} \times (70^\circ\text{C} - 90^\circ\text{C}) \times \frac{1 \text{ kJ}}{10^3 \text{ J}}}_{q_1} + \underbrace{5.0 \text{ g} \times \frac{1 \text{ mol}}{46.07 \text{ g}} \times (-88 \text{ kJ/mol})}_{q_2}$$
$$= -9.68 \text{ kJ}$$



Announcement 3/11

Assignments that are due soon
Pre-Class Assignment Week 6-F

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

1st Chem RePS or SI worksheets and Exam 1 wrapper for Unit 2 will be closed on 11:59 PM 3/11

Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom

SI Session

Saige Douglass – Wed 5:30 – 7:30 PM
at Hashinger 412

Paige Kwan - Thu 5 - 7 PM at ~~Hashinger~~
~~412~~ **KC 134**

SCHMID COLLEGE SCIENCE FORUM

CHEMISTRY IN SPACE: WHAT'S GOING ON UP THERE AND HOW DO WE KNOW?



VINCENT ESPOSITO
Assistant Professor of Chemistry

WEDNESDAY, MARCH 11
12 - 1 p.m. in AF 201

Only 60 years ago, space was thought to be an empty void. Since then, over 350 unique molecules have been detected in space with the help of powerful telescopes, sophisticated laboratory experiments, and advanced computational techniques (termed the "triangle of astrochemistry").

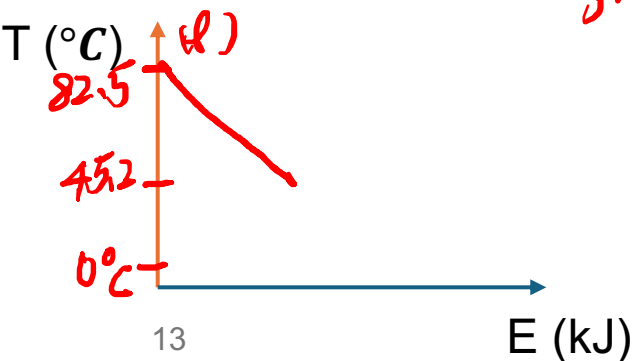
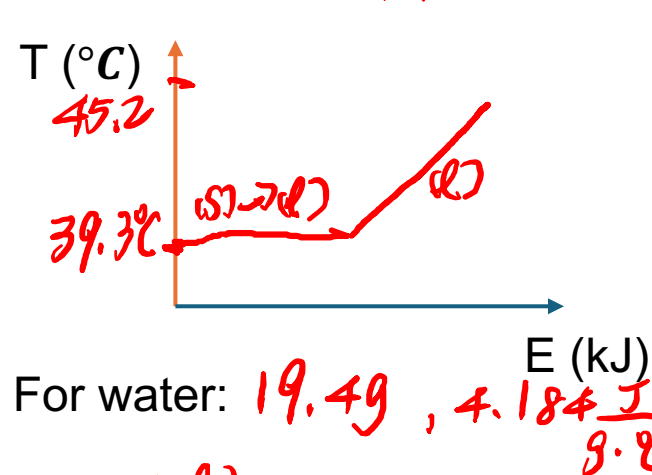
Our understanding of the cosmos grows deeper every day. Dr. Esposito will take you around the Universe to learn about where stars form, how molecules are detected, and how the Esposito Lab contributes to this fascinating process.

Hors d'oeuvres and refreshments will be served!

Your Turn!

A sample of solid C_6H_6 (molar mass = 78.11 g/mol) at its melting point (39.3°C) is placed in 19.4 g water at 82.5°C . All of the solid C_6H_6 melt to liquid and in the end, water and liquid C_6H_6 are under thermal equilibrium with a final temperature of 45.2°C . The heat of fusion of C_6H_6 = 9.92 kJ/mol and the specific heat of liquid C_6H_6 is $1.74 \text{ J/g}^\circ\text{C}$. Specific heat of liquid water is $4.184 \text{ J/g}^\circ\text{C}$ and the freezing point for water is 0°C . What is the mass of the sample of C_6H_6 , in grams?

For C_6H_6 : $78.11 \frac{\text{g}}{\text{mol}}$ $\Delta H = \frac{9.92 \text{ kJ}}{1 \text{ mol}}$ $1.74 \frac{\text{J}}{\text{g}^\circ\text{C}}$ $q(C_6H_6) = -q(H_2O)$



$$q(H_2O) = 4.184 \frac{\text{J}}{\text{g}^\circ\text{C}} \times 19.4 \text{ g} \times (45.2^\circ\text{C} - 82.5^\circ\text{C})$$

$$= -3027.6 \text{ J}$$

$$q(C_6H_6) = m \times \left[\frac{1 \text{ mol}}{78.11 \text{ g}} \times \frac{9.92 \text{ kJ}}{1 \text{ mol}} + m \times 1.74 \frac{\text{J}}{\text{g}^\circ\text{C}} \times (45.2^\circ\text{C} - 39.3^\circ\text{C}) \right]$$

(s) $\rightarrow$ (l) melting

$$= 127 m + 10.266 m = 137 m$$

$$137 m = -(-3027.6 \text{ J})$$

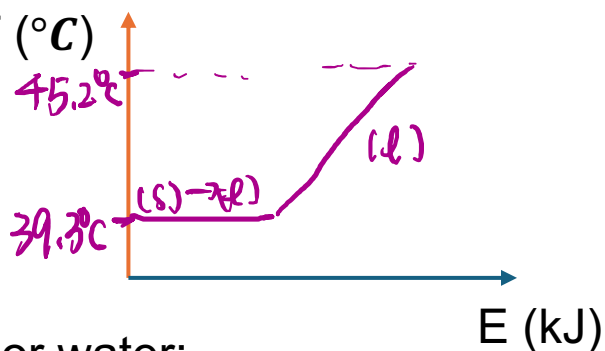
$$m = 22.1 \text{ g}$$

Your Turn!

A sample of solid C_6H_6 (molar mass = 78.11 g/mol) at its melting point (39.3°C) is placed in 19.4 g water at 82.5°C . All of the solid C_6H_6 melt to liquid and in the end, water and liquid C_6H_6 are under thermal equilibrium with a final temperature of 45.2°C . The heat of fusion of C_6H_6 = 9.92 kJ/mol and the specific heat of liquid C_6H_6 is 1.74 J/g $^\circ\text{C}$. Specific heat of liquid water is 4.184 J/g $^\circ\text{C}$ and the freezing point for water is 0°C . What is the mass of the sample of C_6H_6 , in grams?

$$q_{\text{C}_6\text{H}_6} = -q_{\text{H}_2\text{O}}$$

For C_6H_6 :



$$q_{\text{C}_6\text{H}_6} = m \times \frac{1 \text{ mol}}{78.11 \text{ g}} \times \frac{+9.92 \text{ kJ}}{1 \text{ mol}} \times \frac{10^3 \text{ J}}{1 \text{ kJ}} + m \times \frac{1.74 \text{ J}}{1 \text{ g}} \times (45.2^\circ\text{C} - 39.3^\circ\text{C})$$

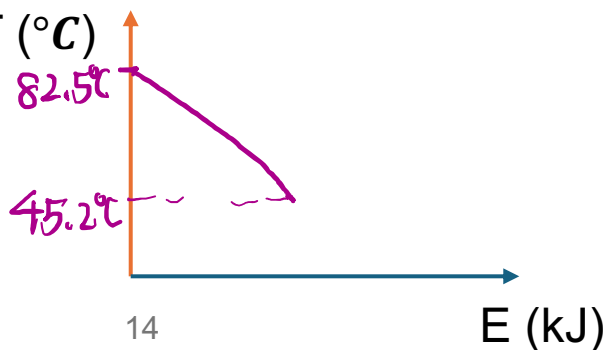
$$= \frac{136.9 \text{ J}}{1 \text{ g}} \times m$$

$$q_{\text{H}_2\text{O}} = 19.4 \text{ g} \times \frac{4.184 \text{ J}}{1 \text{ g}} \times (45.2^\circ\text{C} - 82.5^\circ\text{C}) = -3027.6 \text{ J}$$

$$\frac{136.9 \text{ J}}{1 \text{ g}} \times m = -(-3027.6 \text{ J})$$

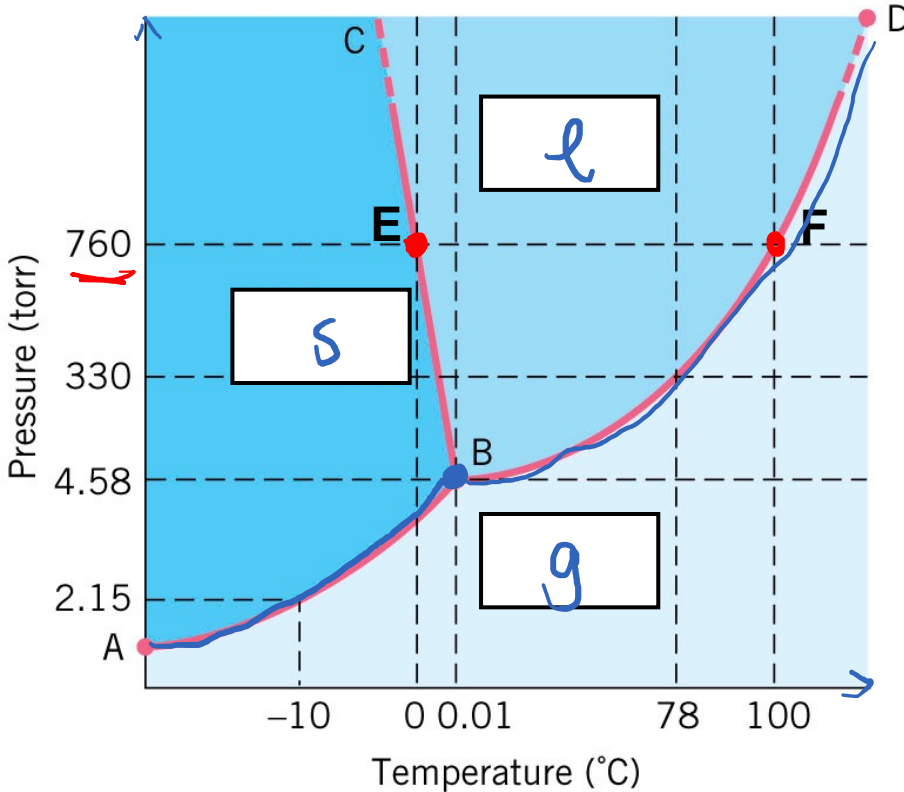
$$m = 22.1 \text{ g}$$

For water:



Phase Diagrams

Phase diagram for H₂O



- The state of matter as a function of pressure (P) and temperature (T) can be summarized as a 2D diagram.

Points:

B = triple point

D = critical point (above it: super critical fluid)

E = normal melting point

F = normal boiling point

Equilibrium lines:

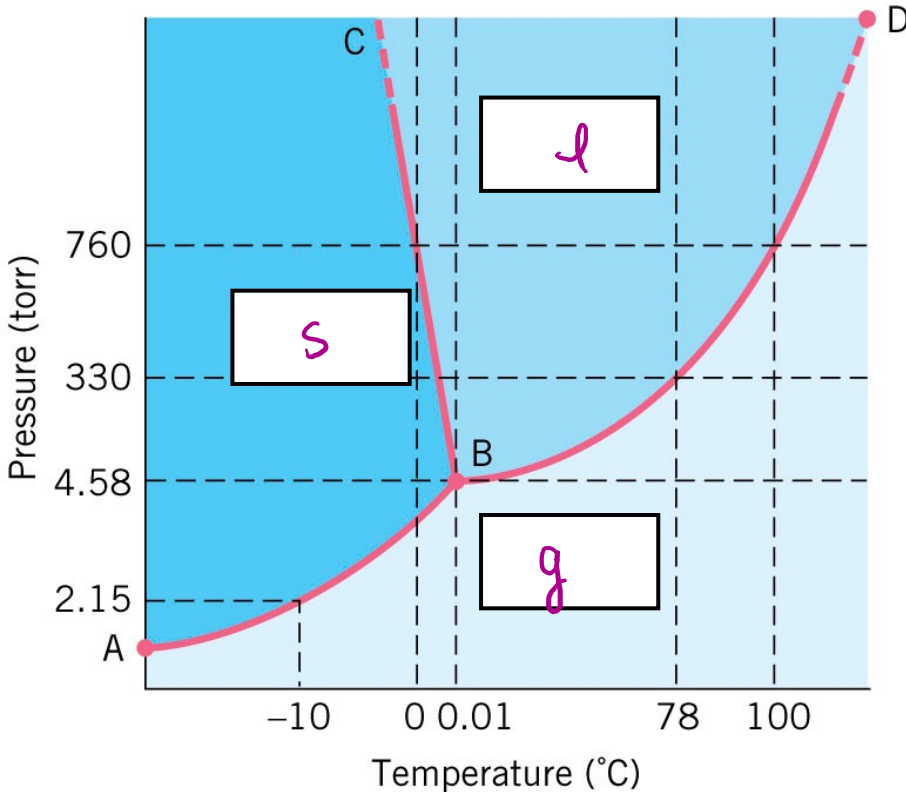
AB = sublimation curve (Above AB is s; Below AB is g)

BD = evaporation curve (Above BD is l; Below BD is g)

BC = melting curve (Left to BC is s; right to BC is l)

Phase Diagrams

Phase diagram for H₂O



- The state of matter as a function of pressure (P) and temperature (T) can be summarized as a 2D diagram.

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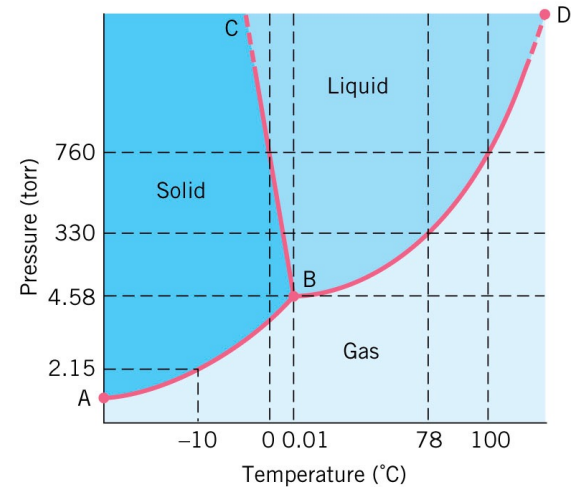
AB = sublimation curve (Above AB is s; Below AB is g)

BD = evaporation curve (Above BD is l; Below BD is g)

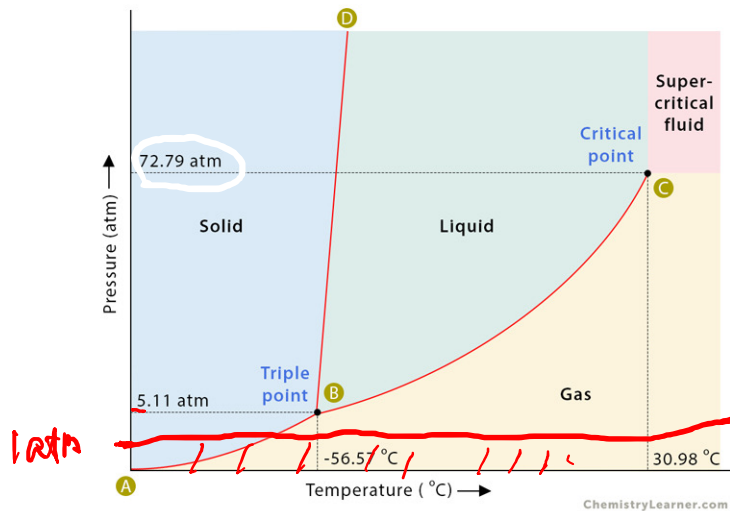
BC = melting curve (Left to BC is s; right to BC is l)



Triple Point of Water 1:18



Phase Diagram of Carbon Dioxide (CO₂)



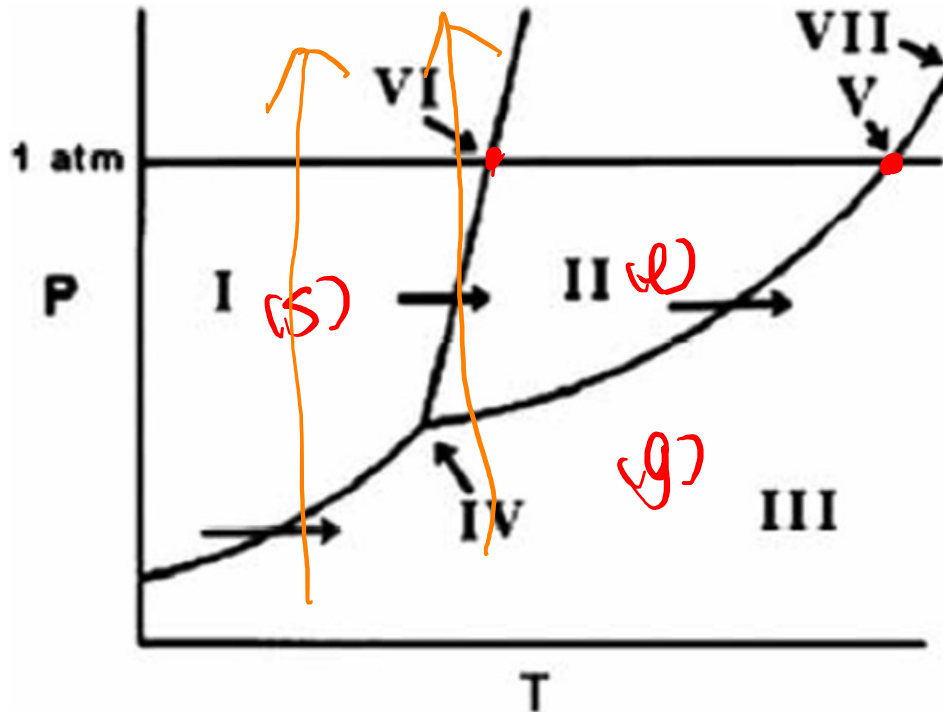
Supercritical Fluid

CO₂ becomes supercritical at ~7.4 MPa and ~31°C.



Your Turn!

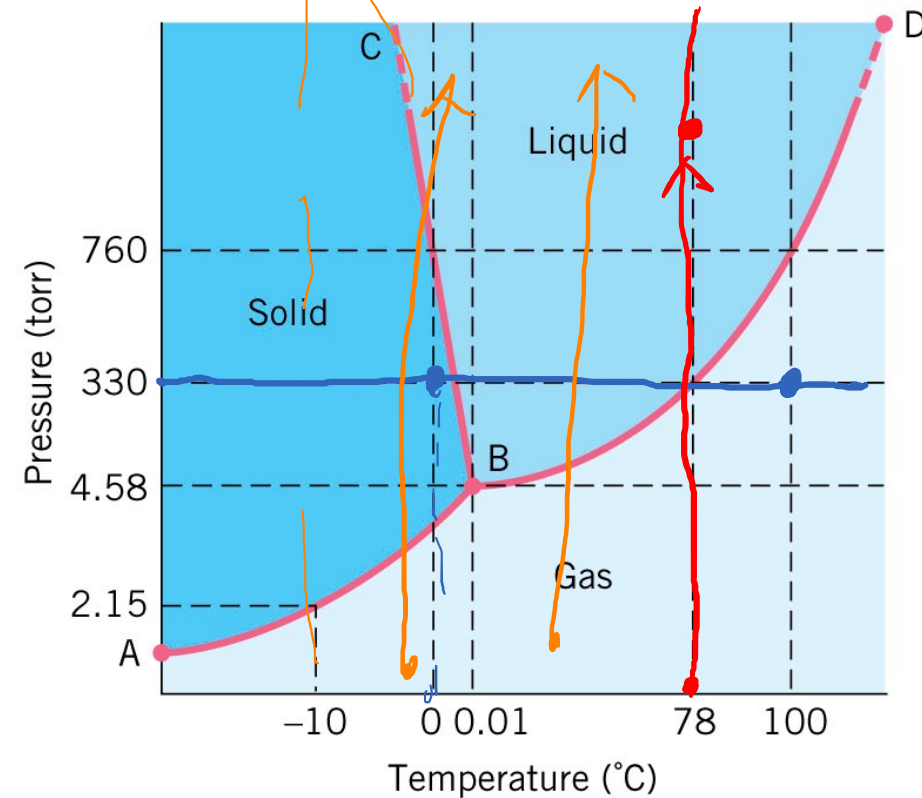
Melting occurs in going from region _____



- a. I to II.
- b. I to III.
- c. II to III.
- d. III to II.
- e. III to I.

Read Phase Diagrams

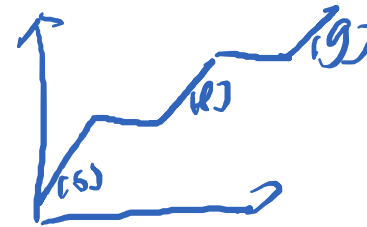
Phase diagram for H₂O



melting curve { left: d(liquid) highest
right: d(solid) highest

Under Constant Pressure:

Draw a line that is parallel to P axis
Ex. Describe the phase changes for a sample of water as the temperature of water increased from 0 °C to 100 °C at 330 torr. (s) → (l) → (g)

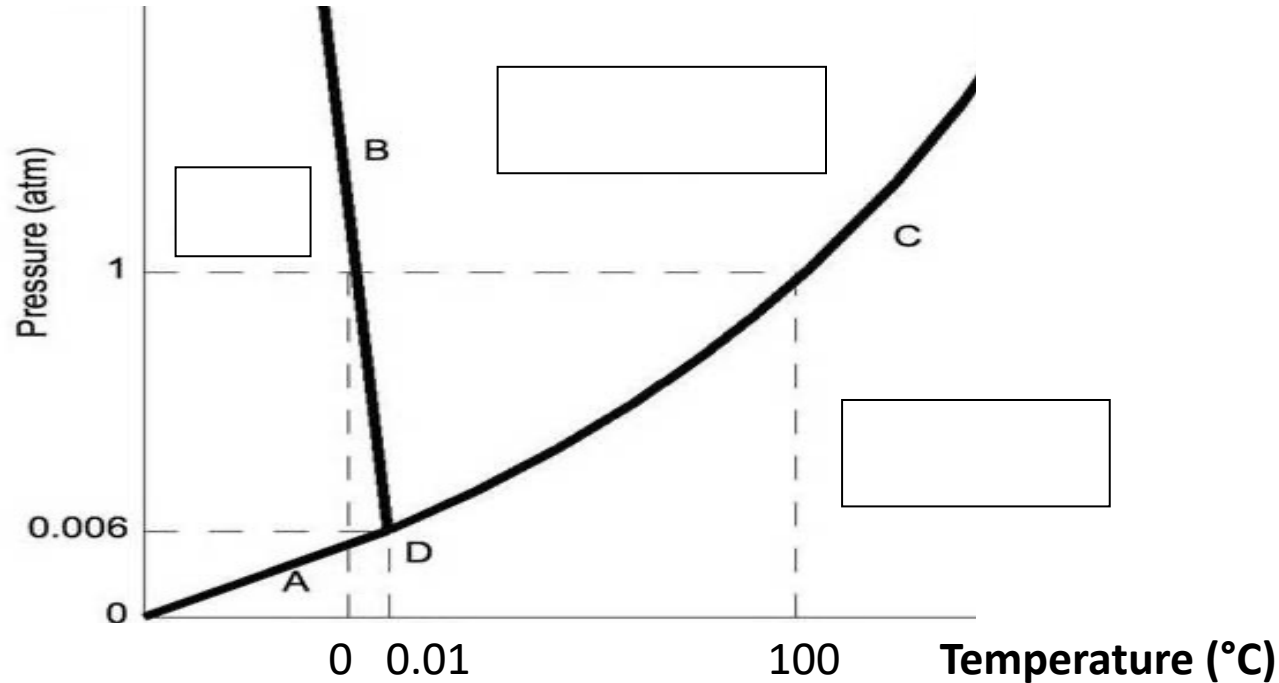


Under Constant temperature:

Draw a line that is parallel to T axis

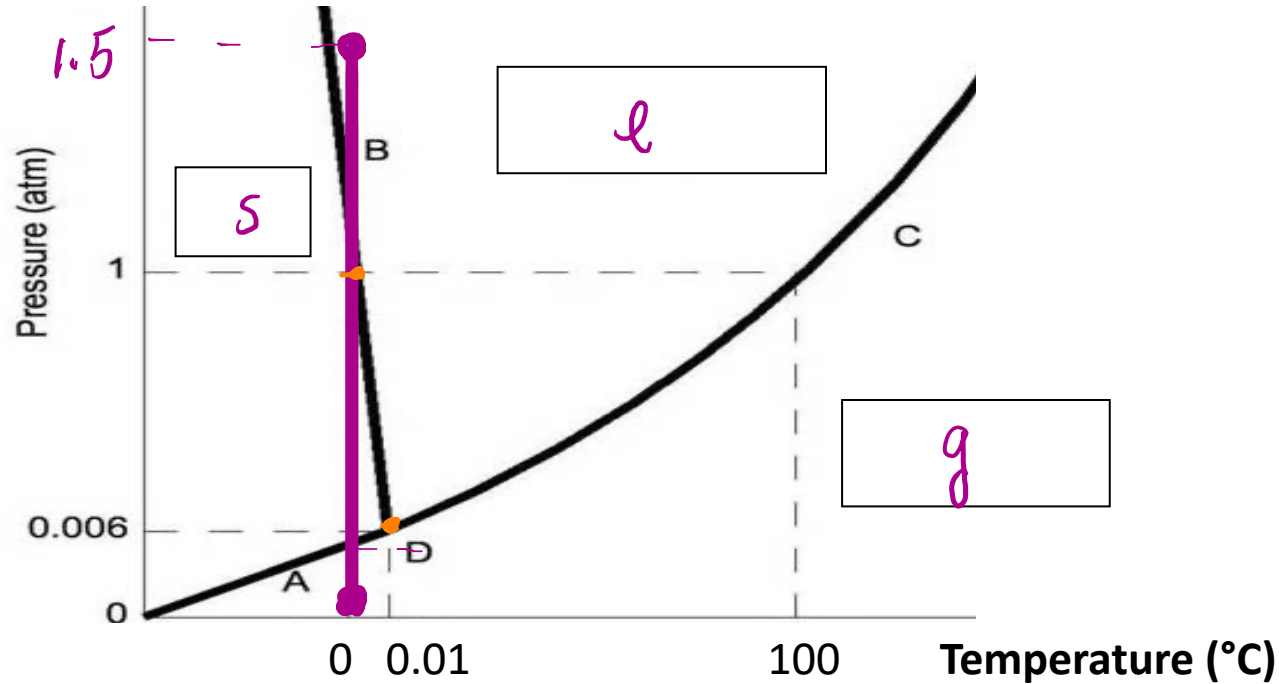
Ex. Describe the phase changes for a sample of water as the pressure of water increased from at 0.0001 torr to 900 torr at 78 °C. (g) → (l)

Your Turn!



- 1) Fill the blank in the diagram.
- 2) What is the phase at 20°C and 0.004 atm ?
- 3) At what temperature and pressure will the three phases of this substance co-exist?
- 4) Describe the phase changes as the pressure of substance increased from 0.00001 atm to 1.5 atm at 0°C .

Your Turn!



1) Fill the blank in the diagram.

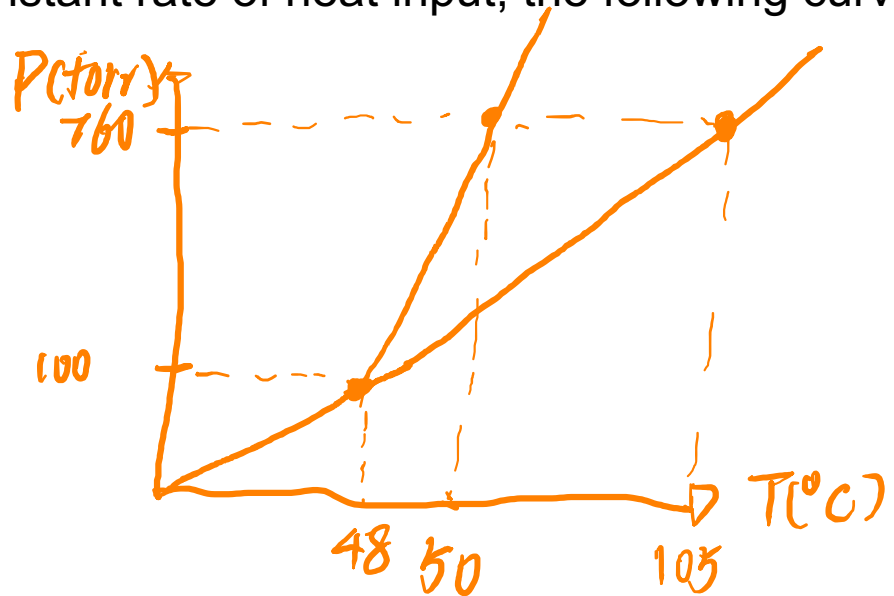
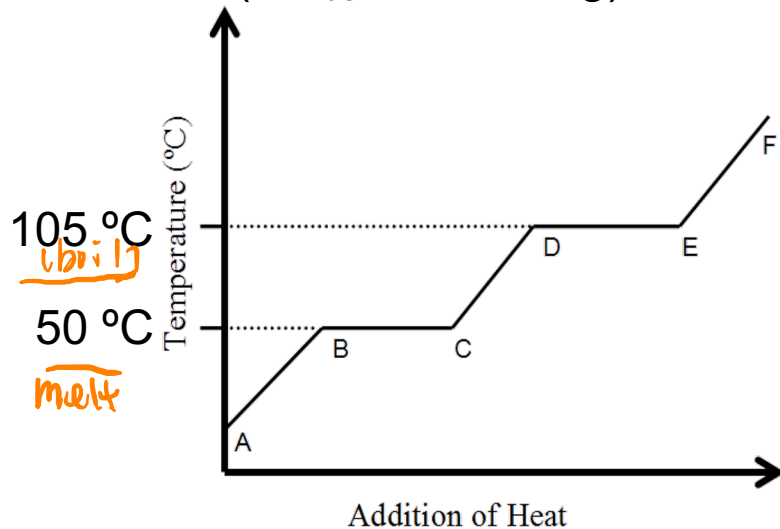
2) What is the phase at 20°C and 0.004 atm? (g)

3) At what temperature and pressure will the three phases of this substance co-exist? 0.006 atm and 0.01 °C

4) Describe the phase changes as the pressure of substance increased from at 0.00001 atm to 1.5 atm at 0 °C . (g) → (s) → (l)

Your Turn!

Substance X has a vapor pressure of 100 torr at its triple point of 48 °C. When 1 mol of substance X is heated at 760 torr with a constant rate of heat input, the following curve is obtained. ($\Delta H_{\text{fus}} = 40.1 \text{ kJ/g}$)



- Sketch the phase diagram on the right.
- At 500 torr and 40° C, what is the phase of substance X?

c. Below what pressure will sublimation occur? _____

d. Which phase is most dense? _____

e. Calculate the amount of heat need to melt 15.5 g of substance X.